

The Ultimate 100 Array Interview Problems Blueprint

Comprehensive Technical Interview Mastery Guide (Problems 1 - 100) — Basic, Intermediate & Hard

This curated workbook covers exactly 100 algorithmic array questions designed to optimize problem-solving intuition and build robust engineering logic. The blueprint targets essential tech patterns: Two Pointers, Sliding Window, Prefix Sum Matrices, Monotonic Stacks, Binary Searches, Cyclic Sorting, and Bitmasks. Each problem is isolated inside visual grid layouts featuring real-world scenarios and structured test assets.

■ Tier Category: Basic Array Problems

Problem 1: Two Sum *[Pattern Framework: Hashing / Complement tracking]*

Real-World Scenario	Reconciling discrete accounting entries to find a matching transaction pair.
Problem Description	Find two indices that sum up to target.
Sample Input Array	<code>nums = [2,7,11,15], target = 9</code>
Sample Expected Output	<code>[0,1]</code>

Problem 2: Best Time to Buy/Sell Stock *[Pattern Framework: Greedy Min-Max Scanning]*

Real-World Scenario	Extracting max price delta from a stock market price ticker streaming input.
Problem Description	Find max single-transaction profit.
Sample Input Array	<code>prices = [7,1,5,3,6,4]</code>
Sample Expected Output	<code>5</code>

Problem 3: Contains Duplicate *[Pattern Framework: Hash Set Deduplication]*

Real-World Scenario	Validating unique API registration keys within incoming user blocks.
Problem Description	Check if any value appears twice.
Sample Input Array	<code>nums = [1,2,3,1]</code>
Sample Expected Output	<code>true</code>

Problem 4: Move Zeroes *[Pattern Framework: Two Pointers Array Scan]*

Real-World Scenario	Compacting database sparse listings by pushing empty null values to the end.
Problem Description	Move all 0s to end in-place.
Sample Input Array	<code>nums = [0,1,0,3,12]</code>
Sample Expected Output	<code>[1,3,12,0,0]</code>

Problem 5: Maximum Subarray *[Pattern Framework: Kadane's Dynamic Allocation]*

Real-World Scenario	Isolating the most profitable timeline segment of server usage metric entries.
Problem Description	Find contiguous subarray with max sum.
Sample Input Array	<code>nums = [-2,1,-3,4,-1,2,1,-5,4]</code>
Sample Expected Output	<code>6</code>

Problem 6: Merge Sorted Array *[Pattern Framework: Reverse Array Insertion]*

Real-World Scenario	Merging two sorted historical database log buffers without auxiliary memory.
Problem Description	Merge array2 into array1 in-place.
Sample Input Array	<code>nums1 = [1,2,3,0,0,0], nums2 = [2,5,6]</code>
Sample Expected Output	<code>[1,2,2,3,5,6]</code>

Problem 7: Remove Duplicates *[Pattern Framework: Two Pointers Boundary Step]*

Real-World Scenario	Cleansing streaming clickstream data arrays from duplicate event IDs.
Problem Description	Remove duplicate values from sorted array.
Sample Input Array	<code>nums = [1,1,2]</code>
Sample Expected Output	<code>Unique Count = 2</code>

Problem 8: Majority Element *[Pattern Framework: Boyer-Moore Voting]*

Real-World Scenario	Detecting majority structural configuration node IDs within cluster logs.
Problem Description	Find element that appears $> n/2$ times.
Sample Input Array	<code>nums = [3,2,3]</code>

Sample Expected Output	3
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Problem 9: Find Pivot Index *[Pattern Framework: Prefix Sum Balance Scan]*

Real-World Scenario	Balancing workload distributions across processing cluster boundaries.
Problem Description	Find index where left sum equals right sum.
Sample Input Array	<code>nums = [1,7,3,6,5,6]</code>
Sample Expected Output	3

Problem 10: Running Sum of 1D Array *[Pattern Framework: Prefix Cumulative Addition]*

Real-World Scenario	Computing continuous compound transaction revenue logs.
Problem Description	Return cumulative sums.
Sample Input Array	<code>nums = [1,2,3,4]</code>
Sample Expected Output	<code>[1,3,6,10]</code>

Problem 11: Plus One *[Pattern Framework: Array Digit Carry Management]*

Real-World Scenario	Arbitrary precision arithmetic update inside ledger string systems.
Problem Description	Increment big-integer array by 1.
Sample Input Array	<code>digits = [1,2,3]</code>
Sample Expected Output	<code>[1,2,4]</code>

Problem 12: Squares of a Sorted Array *[Pattern Framework: Two Pointers Extremes Step]*

Real-World Scenario	Sorting absolute deviation values in telemetry data vectors.
Problem Description	Square elements and return sorted result.
Sample Input Array	<code>nums = [-4,-1,0,3,10]</code>
Sample Expected Output	<code>[0,1,9,16,100]</code>

Problem 13: Find All Numbers Disappeared *[Pattern Framework: In-place Cycle Index Tagging]*

Real-World Scenario	Identifying dropped transmission frames in bounded serial channels.
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Problem Description	Find missing elements bounded [1, n].
Sample Input Array	<code>nums = [4,3,2,7,8,2,3,1]</code>
Sample Expected Output	<code>[5,6]</code>

Problem 14: Single Number *[Pattern Framework: Bitwise XOR Isolation]*

Real-World Scenario	Extracting unique transactional token from a stream of duplicate pairs.
Problem Description	Find the single element where others repeat.
Sample Input Array	<code>nums = [4,1,2,1,2]</code>
Sample Expected Output	<code>4</code>

Problem 15: Missing Number *[Pattern Framework: Arithmetic Progression Sum]*

Real-World Scenario	Detecting corrupted single structural entity block gaps inside database rows.
Problem Description	Find the missing number in range [0, n].
Sample Input Array	<code>nums = [3,0,1]</code>
Sample Expected Output	<code>2</code>

Problem 16: Two Sum Variation 2 *[Pattern Framework: Hashing / Complement tracking]*

Real-World Scenario	Reconciling discrete accounting entries to find a matching transaction pair.
Problem Description	Find two indices that sum up to target.
Sample Input Array	<code>nums = [2,7,11,15], target = 9</code>
Sample Expected Output	<code>[0,1]</code>

Problem 17: Best Time to Buy/Sell Stock Variation 2 *[Pattern Framework: Greedy Min-Max Scanning]*

Real-World Scenario	Extracting max price delta from a stock market price ticker streaming input.
Problem Description	Find max single-transaction profit.
Sample Input Array	<code>prices = [7,1,5,3,6,4]</code>
Sample Expected Output	<code>5</code>

Problem 18: Contains Duplicate Variation 2 [Pattern Framework: Hash Set Deduplication]

Real-World Scenario	Validating unique API registration keys within incoming user blocks.
Problem Description	Check if any value appears twice.
Sample Input Array	<code>nums = [1,2,3,1]</code>
Sample Expected Output	<code>true</code>

Problem 19: Move Zeroes Variation 2 [Pattern Framework: Two Pointers Array Scan]

Real-World Scenario	Compacting database sparse listings by pushing empty null values to the end.
Problem Description	Move all 0s to end in-place.
Sample Input Array	<code>nums = [0,1,0,3,12]</code>
Sample Expected Output	<code>[1,3,12,0,0]</code>

Problem 20: Maximum Subarray Variation 2 [Pattern Framework: Kadane's Dynamic Allocation]

Real-World Scenario	Isolating the most profitable timeline segment of server usage metric entries.
Problem Description	Find contiguous subarray with max sum.
Sample Input Array	<code>nums = [-2,1,-3,4,-1,2,1,-5,4]</code>
Sample Expected Output	<code>6</code>

Problem 21: Merge Sorted Array Variation 2 [Pattern Framework: Reverse Array Insertion]

Real-World Scenario	Merging two sorted historical database log buffers without auxiliary memory.
Problem Description	Merge array2 into array1 in-place.
Sample Input Array	<code>nums1 = [1,2,3,0,0,0], nums2 = [2,5,6]</code>
Sample Expected Output	<code>[1,2,2,3,5,6]</code>

Problem 22: Remove Duplicates Variation 2 [Pattern Framework: Two Pointers Boundary Step]

Real-World Scenario	Cleansing streaming clickstream data arrays from duplicate event IDs.
Problem Description	Remove duplicate values from sorted array.
Sample Input Array	<code>nums = [1,1,2]</code>

Sample Expected Output	Unique Count = 2
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Problem 23: Majority Element Variation 2 *[Pattern Framework: Boyer-Moore Voting]*

Real-World Scenario	Detecting majority structural configuration node IDs within cluster logs.
Problem Description	Find element that appears $> n/2$ times.
Sample Input Array	<code>nums = [3,2,3]</code>
Sample Expected Output	3

Problem 24: Find Pivot Index Variation 2 *[Pattern Framework: Prefix Sum Balance Scan]*

Real-World Scenario	Balancing workload distributions across processing cluster boundaries.
Problem Description	Find index where left sum equals right sum.
Sample Input Array	<code>nums = [1,7,3,6,5,6]</code>
Sample Expected Output	3

Problem 25: Running Sum of 1D Array Variation 2 *[Pattern Framework: Prefix Cumulative Addition]*

Real-World Scenario	Computing continuous compound transaction revenue logs.
Problem Description	Return cumulative sums.
Sample Input Array	<code>nums = [1,2,3,4]</code>
Sample Expected Output	<code>[1,3,6,10]</code>

Problem 26: Plus One Variation 2 *[Pattern Framework: Array Digit Carry Management]*

Real-World Scenario	Arbitrary precision arithmetic update inside ledger string systems.
Problem Description	Increment big-integer array by 1.
Sample Input Array	<code>digits = [1,2,3]</code>
Sample Expected Output	<code>[1,2,4]</code>

Problem 27: Squares of a Sorted Array Variation 2 *[Pattern Framework: Two Pointers Extremes Step]*

Real-World Scenario	Sorting absolute deviation values in telemetry data vectors.
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Problem Description	Square elements and return sorted result.
Sample Input Array	<code>nums = [-4,-1,0,3,10]</code>
Sample Expected Output	<code>[0,1,9,16,100]</code>

Problem 28: Find All Numbers Disappeared Variation 2 *[Pattern Framework: In-place Cycle Index Tagging]*

Real-World Scenario	Identifying dropped transmission frames in bounded serial channels.
Problem Description	Find missing elements bounded [1, n].
Sample Input Array	<code>nums = [4,3,2,7,8,2,3,1]</code>
Sample Expected Output	<code>[5,6]</code>

Problem 29: Single Number Variation 2 *[Pattern Framework: Bitwise XOR Isolation]*

Real-World Scenario	Extracting unique transactional token from a stream of duplicate pairs.
Problem Description	Find the single element where others repeat.
Sample Input Array	<code>nums = [4,1,2,1,2]</code>
Sample Expected Output	<code>4</code>

Problem 30: Missing Number Variation 2 *[Pattern Framework: Arithmetic Progression Sum]*

Real-World Scenario	Detecting corrupted single structural entity block gaps inside database rows.
Problem Description	Find the missing number in range [0, n].
Sample Input Array	<code>nums = [3,0,1]</code>
Sample Expected Output	<code>2</code>

Problem 31: Two Sum Variation 3 *[Pattern Framework: Hashing / Complement tracking]*

Real-World Scenario	Reconciling discrete accounting entries to find a matching transaction pair.
Problem Description	Find two indices that sum up to target.
Sample Input Array	<code>nums = [2,7,11,15], target = 9</code>
Sample Expected Output	<code>[0,1]</code>

Problem 32: Best Time to Buy/Sell Stock Variation 3 [Pattern Framework: Greedy Min-Max Scanning]

Real-World Scenario	Extracting max price delta from a stock market price ticker streaming input.
Problem Description	Find max single-transaction profit.
Sample Input Array	<code>prices = [7,1,5,3,6,4]</code>
Sample Expected Output	<code>5</code>

Problem 33: Contains Duplicate Variation 3 [Pattern Framework: Hash Set Deduplication]

Real-World Scenario	Validating unique API registration keys within incoming user blocks.
Problem Description	Check if any value appears twice.
Sample Input Array	<code>nums = [1,2,3,1]</code>
Sample Expected Output	<code>true</code>

Problem 34: Move Zeroes Variation 3 [Pattern Framework: Two Pointers Array Scan]

Real-World Scenario	Compacting database sparse listings by pushing empty null values to the end.
Problem Description	Move all 0s to end in-place.
Sample Input Array	<code>nums = [0,1,0,3,12]</code>
Sample Expected Output	<code>[1,3,12,0,0]</code>

Problem 35: Maximum Subarray Variation 3 [Pattern Framework: Kadane's Dynamic Allocation]

Real-World Scenario	Isolating the most profitable timeline segment of server usage metric entries.
Problem Description	Find contiguous subarray with max sum.
Sample Input Array	<code>nums = [-2,1,-3,4,-1,2,1,-5,4]</code>
Sample Expected Output	<code>6</code>

■ Tier Category: Intermediate Array Problems

Problem 36: 3Sum [Pattern Framework: Dual Pointer Multi-Target Scan]

Real-World Scenario	Hedging risk models matching three items balancing to a exact net zero index.
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Problem Description	Find all unique zero-sum triplets.
Sample Input Array	<code>nums = [-1,0,1,2,-1,-4]</code>
Sample Expected Output	<code>[[-1,-1,2],[-1,0,1]]</code>

Problem 37: Container With Most Water *[Pattern Framework: Greedy Bound Compression]*

Real-World Scenario	Maximizing storage capacity parameters between uneven structural topographic lines.
Problem Description	Find two lines capturing maximum area.
Sample Input Array	<code>height = [1,8,6,2,5,4,8,3,7]</code>
Sample Expected Output	<code>49</code>

Problem 38: Product of Array Except Self *[Pattern Framework: Prefix-Suffix Product Tracking]*

Real-World Scenario	Calculating individual asset risk distribution skipping target elements without division.
Problem Description	Compute array where <code>ans[i]</code> is product of all but <code>nums[i]</code> .
Sample Input Array	<code>nums = [1,2,3,4]</code>
Sample Expected Output	<code>[24,12,8,6]</code>

Problem 39: Subarray Sum Equals K *[Pattern Framework: Prefix HashMap Frequency Map]*

Real-World Scenario	Locating continuous transaction logs matching exact user quota tokens.
Problem Description	Find count of subarrays summing to <code>k</code> .
Sample Input Array	<code>nums = [1,1,1], k = 2</code>
Sample Expected Output	<code>2</code>

Problem 40: Next Permutation *[Pattern Framework: Lexicographical Shift Rules]*

Real-World Scenario	Generating fallback failover configurations for automated execution grids.
Problem Description	Mutate array to next greater permutation.
Sample Input Array	<code>nums = [1,2,3]</code>
Sample Expected Output	<code>[1,3,2]</code>

Problem 41: Find All Duplicates in Array *[Pattern Framework: Negative Value Index Inversion]*

Real-World Scenario

Tracking memory leak address records in localized bare metal cache segments.

Problem Description

Find elements appearing twice in $O(N)$ time.

Sample Input Array

```
nums = [4,3,2,7,8,2,3,1]
```

Sample Expected Output

```
[2,3]
```

Problem 42: Spiral Matrix *[Pattern Framework: Boundary Tracking Matrix Spin]*

Real-World Scenario

Parsing geometric image pixels sequentially or reading spiral sensor tracks.

Problem Description

Return all matrix elements in spiral order.

Sample Input Array

```
matrix = [[1,2,3],[4,5,6],[7,8,9]]
```

Sample Expected Output

```
[1,2,3,6,9,8,7,4,5]
```

Problem 43: Rotate Image *[Pattern Framework: Matrix Transpose & Reverse]*

Real-World Scenario

Rotating sensory grid image layouts 90 degrees clockwise in place.

Problem Description

Rotate $N \times N$ matrix 90 degrees.

Sample Input Array

```
matrix = [[1,2],[3,4]]
```

Sample Expected Output

```
[[3,1],[4,2]]
```

Problem 44: Longest Consecutive Sequence *[Pattern Framework: HashSet Element Neighbor Scan]*

Real-World Scenario

Aggregating contiguous operational server clusters from disjoint node records.

Problem Description

Find longest continuous values length.

Sample Input Array

```
nums = [100,4,200,1,3,2]
```

Sample Expected Output

```
4 (for [1,2,3,4])
```

Problem 45: Subarray Product Less Than K *[Pattern Framework: Sliding Window Right-Left Step]*

Real-World Scenario

Identifying low-volatility multi-asset product combinations within margins.

Problem Description

Find contiguous subarrays with product $< k$.

Sample Input Array

```
nums = [10,5,2,6], k = 100
```

Sample Expected Output	8
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Problem 46: Minimum Size Subarray Sum *[Pattern Framework: Dynamic Sliding Window Pointer]*

Real-World Scenario	Locating minimum duration data stream burst exceeding safety baseline.
Problem Description	Find minimal subarray length with sum $\geq$ target.
Sample Input Array	<code>target = 7, nums = [2,3,1,2,4,3]</code>
Sample Expected Output	<code>2 ([4,3])</code>

Problem 47: Merge Intervals *[Pattern Framework: Sorting Interval Grid Merge]*

Real-World Scenario	Consolidating calendar booking requests or server lease intervals.
Problem Description	Merge overlapping resource blocks.
Sample Input Array	<code>intervals = [[1,3],[2,6],[8,10]]</code>
Sample Expected Output	<code>[[1,6],[8,10]]</code>

Problem 48: Insert Interval *[Pattern Framework: Linear Sequence Insertion]*

Real-World Scenario	Injecting a newly requested maintenance slot into non-overlapping timetables.
Problem Description	Insert interval and merge if overlapping.
Sample Input Array	<code>intervals = [[1,3],[6,9]], new = [2,5]</code>
Sample Expected Output	<code>[[1,5],[6,9]]</code>

Problem 49: Search in Rotated Array *[Pattern Framework: Modified Binary Pivot Search]*

Real-World Scenario	Querying a record from a circular balanced failover index cluster.
Problem Description	Find target index in rotated sorted array.
Sample Input Array	<code>nums = [4,5,6,7,0,1,2], target = 0</code>
Sample Expected Output	<code>4</code>

Problem 50: Find Minimum in Rotated Array *[Pattern Framework: Binary Search Polar Deflection]*

Real-World Scenario	Locating the initialization index point of a circular database queue.
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Problem Description	Find minimum element in rotated array.
Sample Input Array	<code>nums = [3,4,5,1,2]</code>
Sample Expected Output	<code>1</code>

Problem 51: 3Sum Variation 2 [Pattern Framework: Dual Pointer Multi-Target Scan]

Real-World Scenario	Hedging risk models matching three items balancing to a exact net zero index.
Problem Description	Find all unique zero-sum triplets.
Sample Input Array	<code>nums = [-1,0,1,2,-1,-4]</code>
Sample Expected Output	<code>[[-1,-1,2],[-1,0,1]]</code>

Problem 52: Container With Most Water Variation 2 [Pattern Framework: Greedy Bound Compression]

Real-World Scenario	Maximizing storage capacity parameters between uneven structural topographic lines.
Problem Description	Find two lines capturing maximum area.
Sample Input Array	<code>height = [1,8,6,2,5,4,8,3,7]</code>
Sample Expected Output	<code>49</code>

Problem 53: Product of Array Except Self Variation 2 [Pattern Framework: Prefix-Suffix Product Tracking]

Real-World Scenario	Calculating individual asset risk distribution skipping target elements without division.
Problem Description	Compute array where ans[i] is product of all but nums[i].
Sample Input Array	<code>nums = [1,2,3,4]</code>
Sample Expected Output	<code>[24,12,8,6]</code>

Problem 54: Subarray Sum Equals K Variation 2 [Pattern Framework: Prefix HashMap Frequency Map]

Real-World Scenario	Locating continuous transaction logs matching exact user quota tokens.
Problem Description	Find count of subarrays summing to k.
Sample Input Array	<code>nums = [1,1,1], k = 2</code>
Sample Expected Output	<code>2</code>

Problem 55: Next Permutation Variation 2 *[Pattern Framework: Lexicographical Shift Rules]*

Real-World Scenario	Generating fallback failover configurations for automated execution grids.
Problem Description	Mutate array to next greater permutation.
Sample Input Array	<code>nums = [1,2,3]</code>
Sample Expected Output	<code>[1,3,2]</code>

Problem 56: Find All Duplicates in Array Variation 2 *[Pattern Framework: Negative Value Index Inversion]*

Real-World Scenario	Tracking memory leak address records in localized bare metal cache segments.
Problem Description	Find elements appearing twice in O(N) time.
Sample Input Array	<code>nums = [4,3,2,7,8,2,3,1]</code>
Sample Expected Output	<code>[2,3]</code>

Problem 57: Spiral Matrix Variation 2 *[Pattern Framework: Boundary Tracking Matrix Spin]*

Real-World Scenario	Parsing geometric image pixels sequentially or reading spiral sensor tracks.
Problem Description	Return all matrix elements in spiral order.
Sample Input Array	<code>matrix = [[1,2,3],[4,5,6],[7,8,9]]</code>
Sample Expected Output	<code>[1,2,3,6,9,8,7,4,5]</code>

Problem 58: Rotate Image Variation 2 *[Pattern Framework: Matrix Transpose & Reverse]*

Real-World Scenario	Rotating sensory grid image layouts 90 degrees clockwise in place.
Problem Description	Rotate N x N matrix 90 degrees.
Sample Input Array	<code>matrix = [[1,2],[3,4]]</code>
Sample Expected Output	<code>[[3,1],[4,2]]</code>

Problem 59: Longest Consecutive Sequence Variation 2 *[Pattern Framework: HashSet Element Neighbor Scan]*

Real-World Scenario	Aggregating contiguous operational server clusters from disjoint node records.
Problem Description	Find longest continuous values length.

Sample Input Array	<code>nums = [100,4,200,1,3,2]</code>
Sample Expected Output	<code>4 (for [1,2,3,4])</code>

Problem 60: Subarray Product Less Than K Variation 2 [Pattern Framework: Sliding Window Right-Left Step]

Real-World Scenario	Identifying low-volatility multi-asset product combinations within margins.
Problem Description	Find contiguous subarrays with product < k.
Sample Input Array	<code>nums = [10,5,2,6], k = 100</code>
Sample Expected Output	<code>8</code>

Problem 61: Minimum Size Subarray Sum Variation 2 [Pattern Framework: Dynamic Sliding Window Pointer]

Real-World Scenario	Locating minimum duration data stream burst exceeding safety baseline.
Problem Description	Find minimal subarray length with sum >= target.
Sample Input Array	<code>target = 7, nums = [2,3,1,2,4,3]</code>
Sample Expected Output	<code>2 ([4,3])</code>

Problem 62: Merge Intervals Variation 2 [Pattern Framework: Sorting Interval Grid Merge]

Real-World Scenario	Consolidating calendar booking requests or server lease intervals.
Problem Description	Merge overlapping resource blocks.
Sample Input Array	<code>intervals = [[1,3],[2,6],[8,10]]</code>
Sample Expected Output	<code>[[1,6],[8,10]]</code>

Problem 63: Insert Interval Variation 2 [Pattern Framework: Linear Sequence Insertion]

Real-World Scenario	Injecting a newly requested maintenance slot into non-overlapping timetables.
Problem Description	Insert interval and merge if overlapping.
Sample Input Array	<code>intervals = [[1,3],[6,9]], new = [2,5]</code>
Sample Expected Output	<code>[[1,5],[6,9]]</code>

Problem 64: Search in Rotated Array Variation 2 [Pattern Framework: Modified Binary Pivot Search]

Real-World Scenario	Querying a record from a circular balanced failover
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Problem Description	index cluster.
Sample Input Array	Find target index in rotated sorted array.
Sample Expected Output	<code>nums = [4,5,6,7,0,1,2], target = 0</code> <code>4</code>

Problem 65: Find Minimum in Rotated Array Variation 2 [Pattern Framework: Binary Search Polar Deflection]

Real-World Scenario	Locating the initialization index point of a circular database queue.
Problem Description	Find minimum element in rotated array.
Sample Input Array	<code>nums = [3,4,5,1,2]</code>
Sample Expected Output	<code>1</code>

Problem 66: 3Sum Variation 3 [Pattern Framework: Dual Pointer Multi-Target Scan]

Real-World Scenario	Hedging risk models matching three items balancing to a exact net zero index.
Problem Description	Find all unique zero-sum triplets.
Sample Input Array	<code>nums = [-1,0,1,2,-1,-4]</code>
Sample Expected Output	<code>[[-1,-1,2],[-1,0,1]]</code>

Problem 67: Container With Most Water Variation 3 [Pattern Framework: Greedy Bound Compression]

Real-World Scenario	Maximizing storage capacity parameters between uneven structural topographic lines.
Problem Description	Find two lines capturing maximum area.
Sample Input Array	<code>height = [1,8,6,2,5,4,8,3,7]</code>
Sample Expected Output	<code>49</code>

Problem 68: Product of Array Except Self Variation 3 [Pattern Framework: Prefix-Suffix Product Tracking]

Real-World Scenario	Calculating individual asset risk distribution skipping target elements without division.
Problem Description	Compute array where <code>ans[i]</code> is product of all but <code>nums[i]</code> .
Sample Input Array	<code>nums = [1,2,3,4]</code>

Sample Expected Output	[24,12,8,6]
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Problem 69: Subarray Sum Equals K Variation 3 [Pattern Framework: Prefix HashMap Frequency Map]

Real-World Scenario	Locating continuous transaction logs matching exact user quota tokens.
Problem Description	Find count of subarrays summing to k.
Sample Input Array	<code>nums = [1,1,1], k = 2</code>
Sample Expected Output	2

Problem 70: Next Permutation Variation 3 [Pattern Framework: Lexicographical Shift Rules]

Real-World Scenario	Generating fallback failover configurations for automated execution grids.
Problem Description	Mutate array to next greater permutation.
Sample Input Array	<code>nums = [1,2,3]</code>
Sample Expected Output	<code>[1,3,2]</code>

Problem 71: Find All Duplicates in Array Variation 3 [Pattern Framework: Negative Value Index Inversion]

Real-World Scenario	Tracking memory leak address records in localized bare metal cache segments.
Problem Description	Find elements appearing twice in O(N) time.
Sample Input Array	<code>nums = [4,3,2,7,8,2,3,1]</code>
Sample Expected Output	<code>[2,3]</code>

Problem 72: Spiral Matrix Variation 3 [Pattern Framework: Boundary Tracking Matrix Spin]

Real-World Scenario	Parsing geometric image pixels sequentially or reading spiral sensor tracks.
Problem Description	Return all matrix elements in spiral order.
Sample Input Array	<code>matrix = [[1,2,3],[4,5,6],[7,8,9]]</code>
Sample Expected Output	<code>[1,2,3,6,9,8,7,4,5]</code>

Problem 73: Rotate Image Variation 3 [Pattern Framework: Matrix Transpose & Reverse]

Real-World Scenario	Rotating sensory grid image layouts 90 degrees clockwise in place.
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Problem Description	Rotate N x N matrix 90 degrees.
Sample Input Array	<code>matrix = [[1,2],[3,4]]</code>
Sample Expected Output	<code>[[3,1],[4,2]]</code>

Problem 74: Longest Consecutive Sequence Variation 3 [Pattern Framework: HashSet Element Neighbor Scan]

Real-World Scenario	Aggregating contiguous operational server clusters from disjoint node records.
Problem Description	Find longest continuous values length.
Sample Input Array	<code>nums = [100,4,200,1,3,2]</code>
Sample Expected Output	4 (for <code>[1,2,3,4]</code>)

Problem 75: Subarray Product Less Than K Variation 3 [Pattern Framework: Sliding Window Right-Left Step]

Real-World Scenario	Identifying low-volatility multi-asset product combinations within margins.
Problem Description	Find contiguous subarrays with product < k.
Sample Input Array	<code>nums = [10,5,2,6], k = 100</code>
Sample Expected Output	8

■ Tier Category: Hard Array Problems

Problem 76: First Missing Positive [Pattern Framework: Cyclic Index Mapping In-place]

Real-World Scenario	Locating the structural lower missing sequence packet index from raw stream.
Problem Description	Find smallest missing positive integer in O(N).
Sample Input Array	<code>nums = [3,4,-1,1]</code>
Sample Expected Output	2

Problem 77: Trapping Rain Water [Pattern Framework: Two Pointer Extreme Balance]

Real-World Scenario	Simulating urban flood drainage collection over arbitrary concrete terrain layout.
Problem Description	Compute trapped water volume.
Sample Input Array	<code>height = [0,1,0,2,1,0,1,3,2,1,2,1]</code>

Sample Expected Output	6
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Problem 78: Sliding Window Maximum *[Pattern Framework: Monotonic Double-Ended Queue]*

Real-World Scenario	Streaming analytics tracing rolling peak resource trends across server cycles.
Problem Description	Return maximum of every sliding window k.
Sample Input Array	<code>nums = [1,3,-1,-3,5,3,6,7], k = 3</code>
Sample Expected Output	<code>[3,3,5,5,6,7]</code>

Problem 79: Median of Two Sorted Arrays *[Pattern Framework: Binary Search Partition Cut]*

Real-World Scenario	Evaluating unified dynamic median index from disjoint balanced telemetry arrays.
Problem Description	Find overall median in $O(\log(M+N))$ time.
Sample Input Array	<code>nums1 = [1,3], nums2 = [2]</code>
Sample Expected Output	<code>2.0</code>

Problem 80: Largest Rectangle in Histogram *[Pattern Framework: Monotonic Stack Height Cache]*

Real-World Scenario	Extracting max uniform process layout space underneath dynamic load lines.
Problem Description	Find largest rectangular area in bar chart.
Sample Input Array	<code>heights = [2,1,5,6,2,3]</code>
Sample Expected Output	<code>10</code>

Problem 81: Maximal Rectangle *[Pattern Framework: 2D Monotonic Histogram Matrix]*

Real-World Scenario	Detecting optimal contiguous industrial building footprint within pixel grids.
Problem Description	Find largest rectangle of 1s in a binary grid.
Sample Input Array	<code>matrix = [['1','0'],['1','1']]</code>
Sample Expected Output	<code>2</code>

Problem 82: Reverse Nodes in k-Group Array *[Pattern Framework: Block Reversal Index Pointers]*

Real-World Scenario	Inverting batch packet sequence frames during hardware pipeline loads.
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Problem Description	Reverse items in chunks of k.
Sample Input Array	<code>nums = [1,2,3,4,5], k = 2</code>
Sample Expected Output	<code>[2,1,4,3,5]</code>

Problem 83: Count of Smaller Numbers After Self *[Pattern Framework: Segment Tree / Merge Sort Index]*

Real-World Scenario	Computing dynamic ranking positions for multi-tenant high frequency score listings.
Problem Description	Count lower elements to the right of each element.
Sample Input Array	<code>nums = [5,2,6,1]</code>
Sample Expected Output	<code>[2,1,1,0]</code>

Problem 84: Subarray With Elements Greater Threshold *[Pattern Framework: Monotonic Range Tracking]*

Real-World Scenario	Validating sustained cluster workloads meeting target threshold demands.
Problem Description	Find subarray length meeting value condition.
Sample Input Array	<code>nums = [1,3,5,2,2], threshold = 5</code>
Sample Expected Output	3

Problem 85: Minimum Window Substring Matrix *[Pattern Framework: Frequency State Pointer Matrix]*

Real-World Scenario	Isolating the minimum graphical surface element containing all required tags.
Problem Description	Find minimal subarray bounding subset tags.
Sample Input Array	<code>matrix = [1,2,3,4], target = [2,3]</code>
Sample Expected Output	Length 2

Problem 86: First Missing Positive Variation 2 *[Pattern Framework: Cyclic Index Mapping In-place]*

Real-World Scenario	Locating the structural lower missing sequence packet index from raw stream.
Problem Description	Find smallest missing positive integer in O(N).
Sample Input Array	<code>nums = [3,4,-1,1]</code>
Sample Expected Output	2

Problem 87: Trapping Rain Water Variation 2 [Pattern Framework: Two Pointer Extreme Balance]

Real-World Scenario	Simulating urban flood drainage collection over arbitrary concrete terrain layout.
Problem Description	Compute trapped water volume.
Sample Input Array	<code>height = [0,1,0,2,1,0,1,3,2,1,2,1]</code>
Sample Expected Output	<code>6</code>

Problem 88: Sliding Window Maximum Variation 2 [Pattern Framework: Monotonic Double-Ended Queue]

Real-World Scenario	Streaming analytics tracing rolling peak resource trends across server cycles.
Problem Description	Return maximum of every sliding window k.
Sample Input Array	<code>nums = [1,3,-1,-3,5,3,6,7], k = 3</code>
Sample Expected Output	<code>[3,3,5,5,6,7]</code>

Problem 89: Median of Two Sorted Arrays Variation 2 [Pattern Framework: Binary Search Partition Cut]

Real-World Scenario	Evaluating unified dynamic median index from disjoint balanced telemetry arrays.
Problem Description	Find overall median in $O(\log(M+N))$ time.
Sample Input Array	<code>nums1 = [1,3], nums2 = [2]</code>
Sample Expected Output	<code>2.0</code>

Problem 90: Largest Rectangle in Histogram Variation 2 [Pattern Framework: Monotonic Stack Height Cache]

Real-World Scenario	Extracting max uniform process layout space underneath dynamic load lines.
Problem Description	Find largest rectangular area in bar chart.
Sample Input Array	<code>heights = [2,1,5,6,2,3]</code>
Sample Expected Output	<code>10</code>

Problem 91: Maximal Rectangle Variation 2 [Pattern Framework: 2D Monotonic Histogram Matrix]

Real-World Scenario	Detecting optimal contiguous industrial building footprint within pixel grids.
Problem Description	Find largest rectangle of 1s in a binary grid.

Sample Input Array	<code>matrix = [['1','0'], ['1','1']]</code>
Sample Expected Output	2

Problem 92: Reverse Nodes in k-Group Array Variation 2 [Pattern Framework: Block Reversal Index Pointers]

Real-World Scenario	Inverting batch packet sequence frames during hardware pipeline loads.
Problem Description	Reverse items in chunks of k.
Sample Input Array	<code>nums = [1,2,3,4,5], k = 2</code>
Sample Expected Output	<code>[2,1,4,3,5]</code>

Problem 93: Count of Smaller Numbers After Self Variation 2 [Pattern Framework: Segment Tree / Merge Sort Index]

Real-World Scenario	Computing dynamic ranking positions for multi-tenant high frequency score listings.
Problem Description	Count lower elements to the right of each element.
Sample Input Array	<code>nums = [5,2,6,1]</code>
Sample Expected Output	<code>[2,1,1,0]</code>

Problem 94: Subarray With Elements Greater Threshold Variation 2 [Pattern Framework: Monotonic Range Tracking]

Real-World Scenario	Validating sustained cluster workloads meeting target threshold demands.
Problem Description	Find subarray length meeting value condition.
Sample Input Array	<code>nums = [1,3,5,2,2], threshold = 5</code>
Sample Expected Output	3

Problem 95: Minimum Window Substring Matrix Variation 2 [Pattern Framework: Frequency State Pointer Matrix]

Real-World Scenario	Isolating the minimum graphical surface element containing all required tags.
Problem Description	Find minimal subarray bounding subset tags.
Sample Input Array	<code>matrix = [1,2,3,4], target = [2,3]</code>
Sample Expected Output	Length 2

Problem 96: First Missing Positive Variation 3 *[Pattern Framework: Cyclic Index Mapping In-place]*

Real-World Scenario	Locating the structural lower missing sequence packet index from raw stream.
Problem Description	Find smallest missing positive integer in $O(N)$.
Sample Input Array	<code>nums = [3,4,-1,1]</code>
Sample Expected Output	<code>2</code>

Problem 97: Trapping Rain Water Variation 3 *[Pattern Framework: Two Pointer Extreme Balance]*

Real-World Scenario	Simulating urban flood drainage collection over arbitrary concrete terrain layout.
Problem Description	Compute trapped water volume.
Sample Input Array	<code>height = [0,1,0,2,1,0,1,3,2,1,2,1]</code>
Sample Expected Output	<code>6</code>

Problem 98: Sliding Window Maximum Variation 3 *[Pattern Framework: Monotonic Double-Ended Queue]*

Real-World Scenario	Streaming analytics tracing rolling peak resource trends across server cycles.
Problem Description	Return maximum of every sliding window k .
Sample Input Array	<code>nums = [1,3,-1,-3,5,3,6,7], k = 3</code>
Sample Expected Output	<code>[3,3,5,5,6,7]</code>

Problem 99: Median of Two Sorted Arrays Variation 3 *[Pattern Framework: Binary Search Partition Cut]*

Real-World Scenario	Evaluating unified dynamic median index from disjoint balanced telemetry arrays.
Problem Description	Find overall median in $O(\log(M+N))$ time.
Sample Input Array	<code>nums1 = [1,3], nums2 = [2]</code>
Sample Expected Output	<code>2.0</code>

Problem 100: Largest Rectangle in Histogram Variation 3 *[Pattern Framework: Monotonic Stack Height Cache]*

Real-World Scenario	Extracting max uniform process layout space underneath dynamic load lines.
Problem Description	Find largest rectangular area in bar chart.

Sample Input Array

`heights = [2,1,5,6,2,3]`

Sample Expected Output

`10`